Edutech

PEC HACKS 3.0

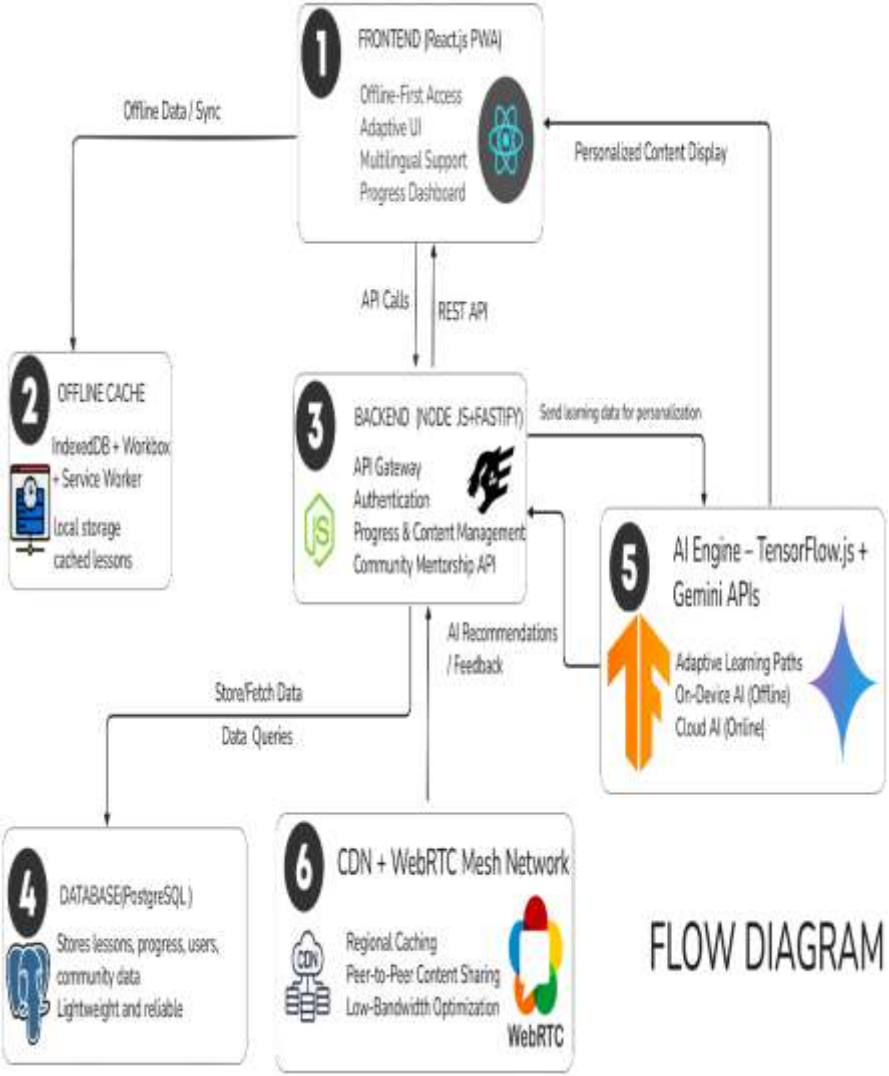
# RURALLEARN THUNDER

PEC HACKS

# Problem Statement

## Personalized Learning App For Rural Students

Many students in rural areas face limited access to quality education due to poor internet, low-spec devices, and fewer qualified teachers. Available learning materials are mostly one-size-fits-all, failing to adapt to each student’s pace or needs.
As a result, rural students struggle to keep up, miss personalized guidance, and risk falling behind academically




# Solution

RuralLearn is a personalized, offline-first learning platform that adapts to each student's pace, device, and connectivity. It delivers locally relevant content, assessments, and guidance to ensure effective learning for every rural student

**01 | Offline-First Access:** Learn anytime, even without internet

**02 | Adaptive Learning Paths:** Content adjusts to student performance and pace.

**03 | Multilingual Support:** Lessons in local languages with cultural context.

**04 | Progress Tracking & Assessments:** Personalized feedback and exam prep.

**05 | Community Mentorship:** Teachers, parents, or peers can support learning.

# Dependencies and Showstopper

1. RuralLearn relies on offline-first web technologies like Progressive Web Apps and Service Workers. Local storage via IndexedDB ensures learning continues without internet.AI-driven personalization uses lightweight TensorFlow.js models locally. When online, cloud APIs enhance content generation and adaptive learning

2. The platform must support low-spec smartphones, tablets, and basic computers. Content is optimized for low bandwidth and intermittent network availability. Background sync ensures progress is updated once connectivity resumes. Performance optimization is crucial to prevent app crashes on constrained devices.

3. Access to accurate regional curriculum data is essential for relevance. Multilingual lessons and cultural adaptations improve engagement. Mentors, teachers, or parents provide guidance, feedback, and content contribution. Community support helps overcome the absence of formal educational resources

4. Extended internet outages may limit cloud AI features and updates. Limited storage can prevent offline content from being downloaded. Missing curriculum data or local content reduces learning effectiveness. Low adoption by students, mentors, or communities can block the platform’s impact

# TECH STACK

## Frontend

React.js (PWA) for responsive and offline-first web experience, with TypeScript for better maintainability and scalability

## Mobile App

React Native for Android compatibility, allowing lightweight installation and cross-platform learning access

## Backend

Node.js with Fastify framework for efficient API handling,plus SQLite/PostgreSQL for reliable and lightweight data storage

## AI & Personalization:

TensorFlow.js for on-device AI processing, and OpenAI or Gemini APIs for advanced personalization when online.

## Offline & Sync Layer

IndexedDB for Local data storage, Service Workers and Workbox for caching and background

## Infrastructure & Network:

CDN with regional edge servers and WebRTC mesh sharing, to support low-bandwidth rural environments efficiently

# THUNDER

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